

Analytical Study of a Wholesale and Retail Marble, Tiles, and Sanitary Shop

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project titled “**Analytical Study of a Wholesale and Retail Marble, Tiles, and Sanitary Shop**”. I extend my appreciation to **Maa Vindhya wasini Marble**, for providing the necessary resources that enabled me to conduct my project.

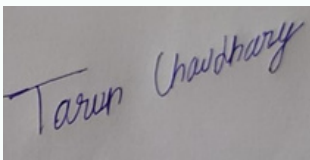
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate
Name: Tarun Chaudhary
Date: 10/02/2026

Executive Summary: Maa Vindhyawasini Marble

The project is focused on studying and analysing the challenges faced by Maa Vindhyawasini Marble, a wholesale and retail dealer of marble, tiles, and sanitary ware located in Ghazipur, Uttar Pradesh. The business has been operating for more than two decades and serves both individual home builders and contractors in the local market.

In recent years, the owner has observed fluctuations in sales and increasing pricing pressure due to competition. Since the business involves high-value inventory and bulk purchases, even small gaps between purchase and sales trends can affect overall profitability. However, there is no structured analysis of past financial performance to clearly understand whether revenue is stable, declining, or seasonal in nature.

To address this, I plan to analyse five years of historical sales and purchase data to understand revenue patterns, margin trends, and inventory movement. Through this analysis, the aim is to identify whether capital is being efficiently utilised and to provide practical recommendations based on actual financial trends of the business.



[Maa Vindhyawasini Marble, Paraspura, Gazipur, UP]

Organisation Background

Business Name: **Maa Vindhyaasini Marble**

Address: **Paraspura, near Power House, Ghazipur, Uttar Pradesh**

Owner's Name: **Mrs. Anju Saraf**

Maa Vindhyaasini Marble is a wholesale and retail dealership of marble, tiles, and sanitary ware with a market presence of more than 20 years in Ghazipur. The business operates from Paraspura near the Power House area and serves both individual home builders and contractors.

It is an authorized dealer of brands such as Varmora.

The showroom deals in branded tiles and sanitary products and is associated with companies such as Varmora. Over time, the business has built strong relationships with contractors, local builders, and repeat customers in the region. It operates with a team of around 20–25 employees who handle sales, inventory, logistics, and daily operations.

Originally established as a family-run enterprise, the business is now managed by the second generation. Due to its long-standing presence in the market, it has developed trust and recognition among customers and suppliers in the local area.

Problem Statement

1) Revenue and Margin Uncertainty

The business owner does not have a structured understanding of long-term sales trends and margin movement. Although monthly sales occur regularly, there is uncertainty about whether revenue is stable, seasonal, or declining. Pricing and margin decisions are mostly experience-based rather than data-driven.

2) Inventory and Capital Efficiency

Since the business involves high-value inventory, inefficient alignment between purchases and sales can lead to capital being locked in stock. There is no systematic evaluation of inventory turnover to assess whether capital is being utilized efficiently.

Background of the Problem

The challenges faced by Maa Vindhya wasini Marble arise mainly from the nature of the tiles and sanitary business, which involves high-value products, bulk purchases, and competitive pricing. Although the business has been operating for many years, there has been no structured analysis of past sales and purchase records to clearly understand revenue patterns and margin movement.

Over time, increased competition and pricing pressure in the local market have made it difficult to judge whether sales fluctuations are temporary or part of a long-term trend. Since tiles and sanitary products require significant capital investment, even small inefficiencies in purchasing or slow-moving stock can affect overall profitability.

In summary, the core issue is not lack of sales, but the absence of systematic financial analysis to understand revenue stability and capital utilization. Without this clarity, it becomes difficult to assess business performance accurately and make informed operational decisions.

Problem Solving Approach

a. Methods Used with Justification

To understand the problems related to revenue stability and inventory efficiency, I will use both quantitative and qualitative methods.

Quantitative Methods

Time-Series Analysis:

Since I have five years of monthly sales and purchase data, time-series analysis will help identify trends, growth patterns, seasonal fluctuations, and possible decline in revenue over time.

Ratio and Margin Analysis:

By calculating gross margin, purchase-to-sales ratio, and approximate inventory turnover, I will evaluate whether profitability and capital utilization have changed over the years.

These methods are suitable because the main problems relate to financial performance and can be measured numerically.

Qualitative Methods

Owner Interaction and Market Observation:

I have discussed with the owner to understand pricing decisions, purchase practices, and market challenges. Observing competitor pricing and customer behavior will support the interpretation of financial results.

This helps connect numerical findings with real business conditions.

b. Intended Data Collection with Justification

Sales Data (5 Years):

Monthly sales records will be collected to analyze revenue growth, volatility, and seasonality.

Purchase Data (5 Years):

Monthly purchase records will help evaluate inventory movement and capital investment in stock.

These variables are important because comparing sales and purchases allows assessment of margin trends and inventory efficiency without requiring complex or unavailable data.

c. Analysis Tools with Justification

Microsoft Excel / Google Sheets:

Used for data cleaning, organizing monthly records, and basic calculations such as averages, growth rates, and ratios.

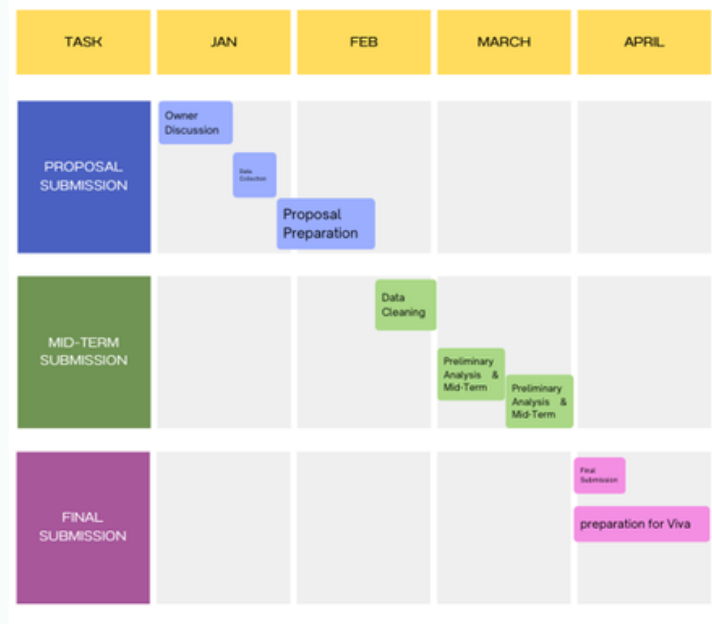
Python (Pandas, Matplotlib, Seaborn):

Used for structured data formatting, deeper analysis, and visualizing long-term trends more clearly.

Excel helps in basic computation and structured presentation, while Python supports better data handling and visualization for five years of records.

Expected Timeline

- Discussion with Shop Owner: 1st Jan – 17th Jan 2026
- Data Collection (Sales & Purchase Records): 18th Jan – 20th Jan 2026
- Proposal Submission: 13 February 2026
- Data Cleaning and Structuring: 13th Feb – 28th Feb 2026
- Preliminary Analysis and Mid-Term Submission: 1st Week of March 2026
- Detailed Financial and Inventory Analysis: Entire March
- Report Drafting, Review, and Final Submission: April 2026



Expected Outcome

- To clearly understand the five-year trend in sales and identify whether revenue is stable, seasonal, or declining.
- To estimate margin patterns over time by comparing sales and purchase data, and observe whether profitability is improving or compressing.
- To evaluate the relationship between purchases and sales in order to understand inventory movement and capital utilization efficiency.
- To provide the owner with a structured financial overview of the business based on actual historical data rather than assumptions.
- To support better purchasing and pricing decisions through data-based insights.